OMNI CAMERA ROVER | REV B PROTOTYPE

1. Controller

File: controller.kicad_sch

2. Motors

File: motors.kicad_sch

3. Charging

File: charging.kicad_sch

4. Regulator

File: regulator.kicad_sch

5. Interlock

File: interlock.kicad_sch

Battery protection

File: battery_protection.kicad_sch

PWR_FLAG GND

PWR_FLAG VMOTOR

PWR_FLAG USB_5V

PWR_FLAG U2_VINT

PWR_FLAG U3_VINT

Editable functional sheets. Global net names link sheets.
Prototype: electrical, thermal and mechanical qualification remains open.

Nathaniel Sanford

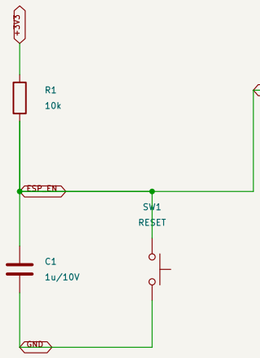
Sheet: /
File: omni_ugv.kicad_sch

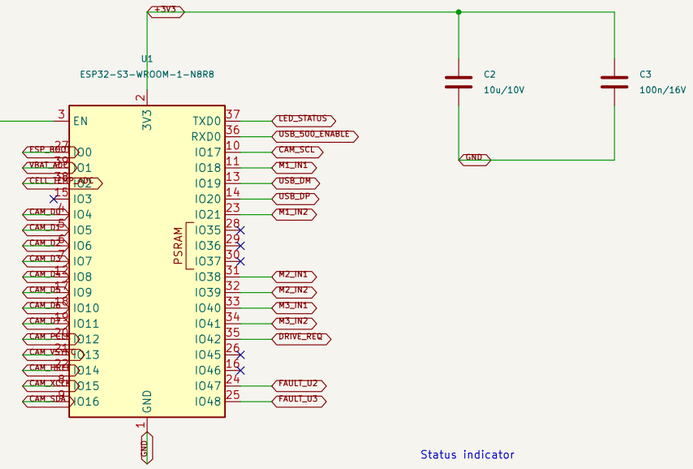
Title: Omni UGV / Rev B

Size: A3	Date: 2026-11-02	**Rev: B-PROTOTYPE**
KiCad E.D.A. 10.0.8		Id: 1/7

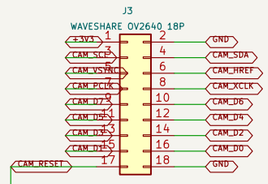
Controller, camera and service points

RESET / enable RC





Waveshare OV2640 | documented 2 x 9 interface



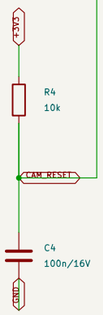
Status indicator

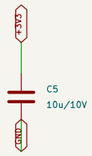


BOOT strap



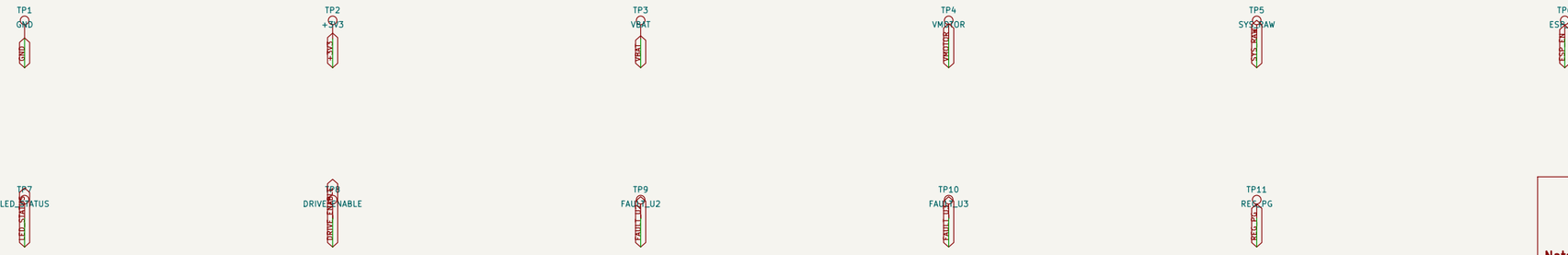
Camera reset RC





[SPEC] GPIO35 / 36 / 37 reserved for N8R8 PSRAM.

Service points (named nets continue on other sheets)



Nate | Omni camera rover

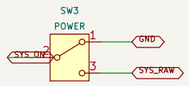
Sheet: /1. Controller/
File: controller.kicad_sch

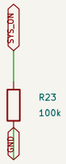
Title: Controller, camera and service points

Size: A3	Date:	**Rev: B — prototype**
KiCad E.D.A. 10.0.8		Id: 2/7

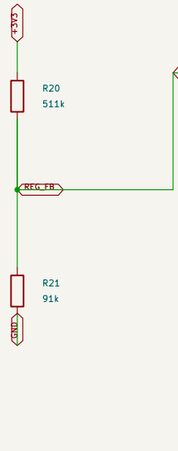
3.3 V buck–boost supply

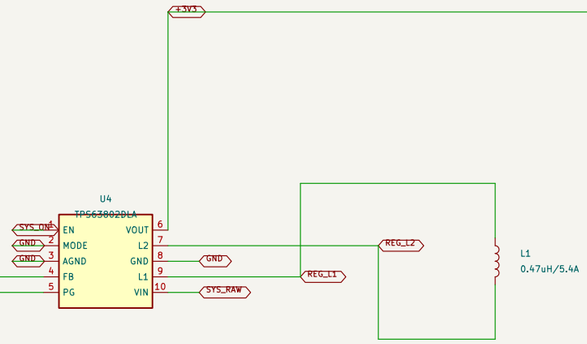
System enable



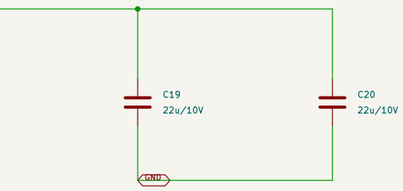


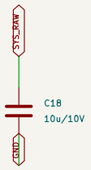
Feedback divider





Output capacitors



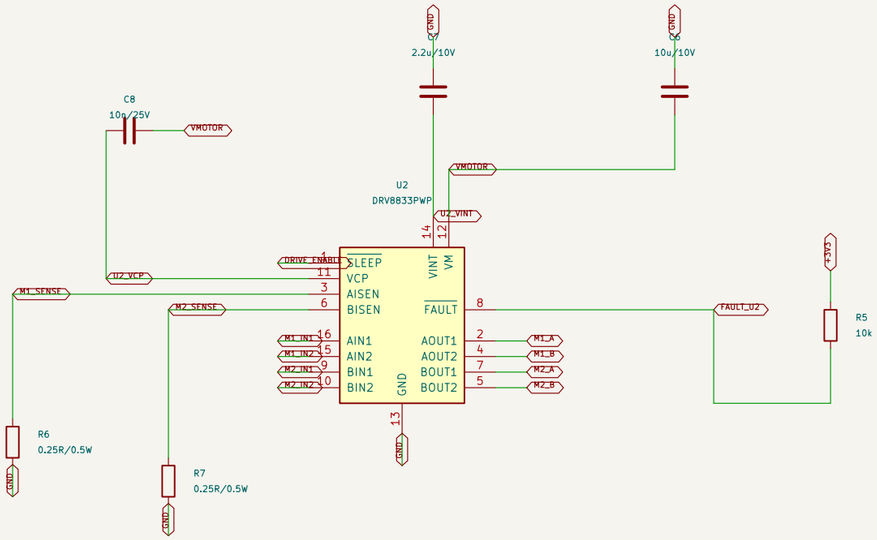


[TARGET] Keep switching loops short. Place input/output capacitors at U4.
[TBD–MEASURE] Verify startup, load transients and capacitor effective capacitance.

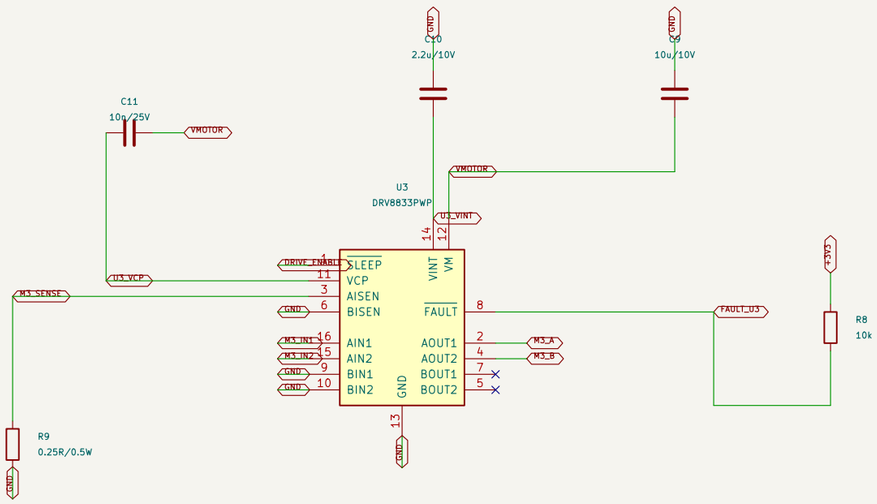
Nate | Omni camera rover		
Sheet: /4. Regulator/ File: regulator.kicad_sch		
Title: 3.3 V buck–boost supply		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 3/7

Three independent motor channels

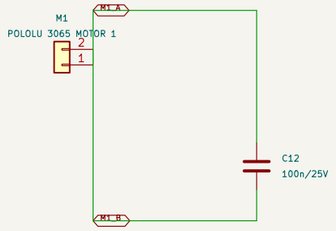
U2: wheels 1 and 2

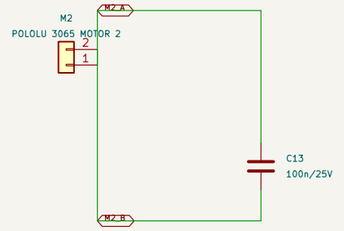


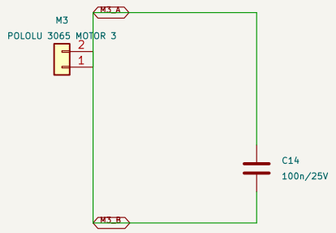
U3: wheel 3; unused bridge pins retained as designed

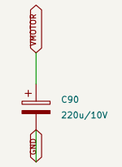


Motor terminal pads and suppression capacitors









[TARGET] Route motor current away from logic returns.
[TBD-MEASURE] Braking overshoot, driver temperature and carpet load.

Nate | Omni camera rover

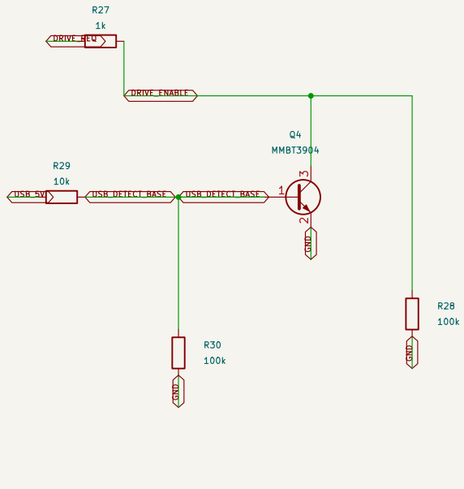
Sheet: /2. Motors/
File: motors.kicad_sch

Title: Three independent motor channels

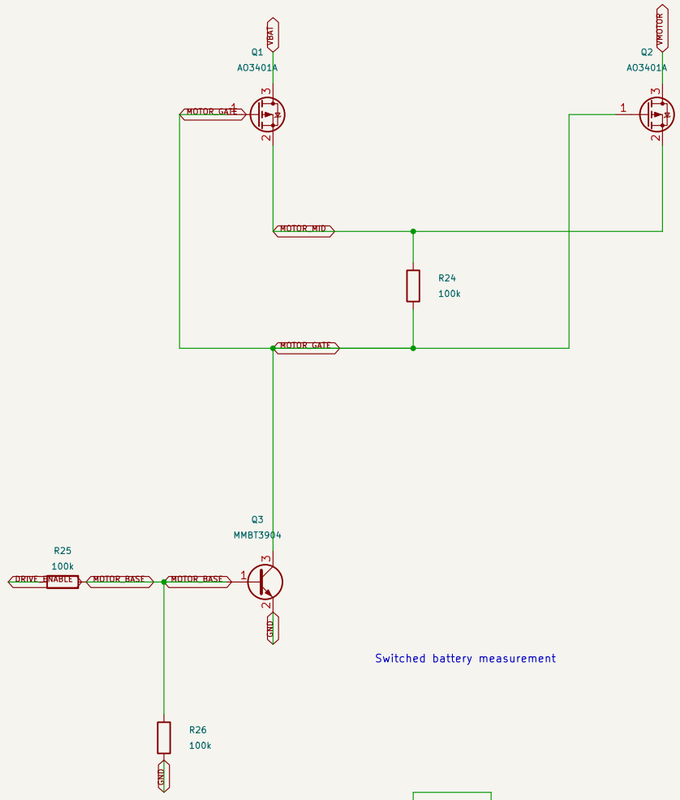
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 4/7

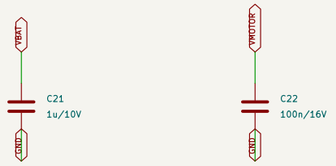
USB motor interlock and battery measurement

USB presence clamps DRIVE_ENABLE

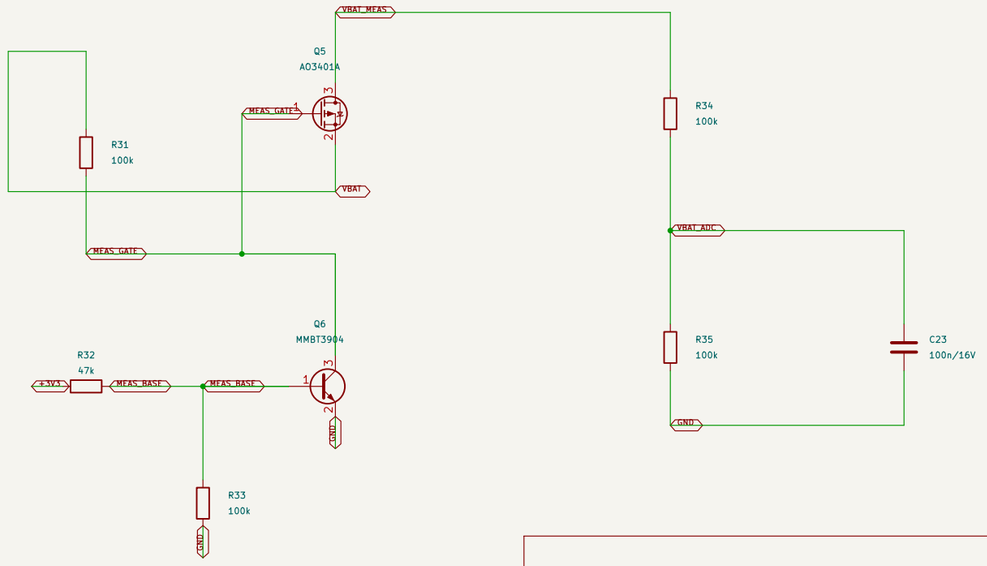


Back-to-back motor supply switch





Switched battery measurement

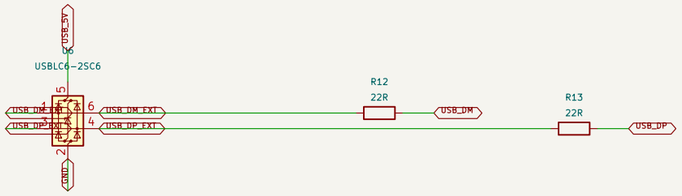


[TARGET] Reset / absent commands keep motors disabled.
[TBD-MEASURE] Verify USB insertion/removal and motor braking transients.

Nate | Omni camera rover		
Sheet: /5. Interlock/		
File: interlock.kicad_sch		
Title: USB motor interlock and battery measurement		
Size: A3	Date:	**Rev: B – prototype**
KiCad E.D.A. 10.0.8		Id: 5/7

USB input, charging and cell connections

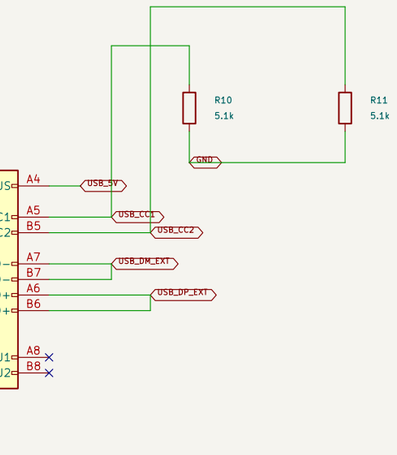
USB data to ESP32 native USB

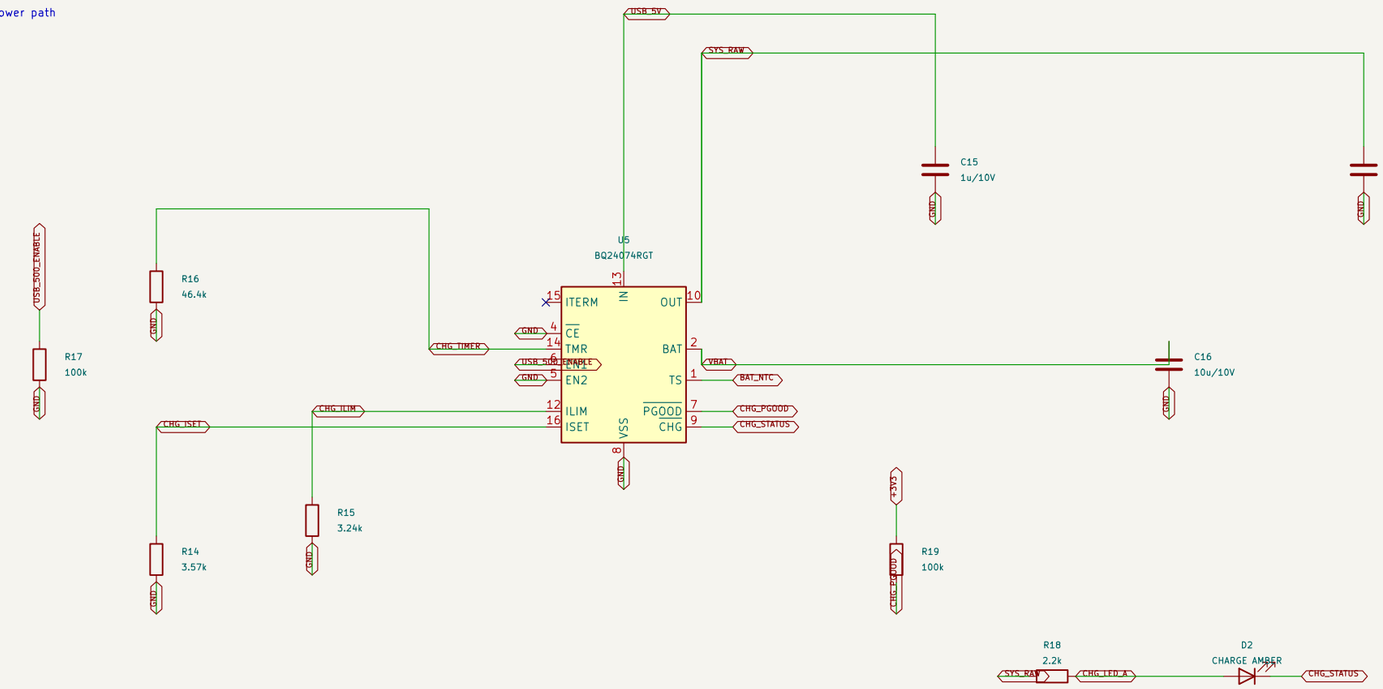


Raw cell –> protection sheet

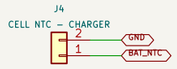


Charge / system power path





Cell-bonded charger probe



[SPEC] J2 CELL_N is not system GND: R36 is the only intended return path.
J4: independent insulated two-wire 103AT-2 probe on the cell.
Motor current uses protected VBAT, not charger SYS_RAW output.

Nate | Omni camera rover	
Sheet: /3. Charging/	
File: charging.kicad_sch	
Title: USB input, charging and cell connections	
Size: A3	Date:
KiCad E.D.A. 10.0.8	**Rev: B – prototype**
	Id: 6/7

1S protection and cell-temperature monitoring

Cell voltage / current sensing

R37
100R 1%
CELL_P
PROT_VDD

R38
20R 1%
CELL_P
PROT_BAT
PROT_VDD

U7
BQ298217RUGR
1 BAT
2 VDD
3 VSS
4 CS
5 CTR
6 PACK
7 DSG
8 CHG
CELL_N
GND
CELL_N
PROT_PACK
PROT_DSG
PROT_CHG

C24
100n/16V X7R
CELL_N

R39
1k 1%
VBAT
PROT_PACK

R36
10m 1% 1W
CELL_N
GND

Common-drain high-side protection FET pair

R41
10M
VBAT

Q8
CSD17308Q3
PROT_DSG
G
D_[5-9]
S_[1-3]
5
1
VBAT

Q7
CSD17308Q3
CELL_P
S_[1-3]
D_[5-9]
5
PROT_DRAIN
G
4
PROT_CHG

R40
10M
CELL_P

TH2 -> J5 -> GPIO1 / ADC1

+3V3

R42
10k 1%

J5
CELL NTC - DRIVE
2
1
GND
CELL_TEMP_DIV

CELL_TEMP_DIV

R43
1k 1%
CELL_TEMP_ADC

TH2
103AT-2 CELL-MOUNTED
GND

C25
100n/16V X7R

TH1 -> J4 charger input

BAT_NTC

TH1
103AT-2 CELL-MOUNTED

GND

#FLG20
PROTECTION_PWR_FLAG
PWR
CELL_N

#FLG21
PROTECTION_PWR_FLAG
PWR
PROT_VDD

[SPEC] TH1 and TH2 are OFF-BOARD, bonded to the cell through electrical insulation.
Separate two-wire probes; do not connect either sensor to raw CELL_N.
[TARGET] Firmware must stop driving on overtemperature, open/short probe or stale readings.

Nate | Omni camera rover	
Sheet: /Battery protection/	
File: battery_protection.kicad_sch	
Title: 1S protection and cell-temperature monitoring	
Size: A3 | Date:	**Rev: B — prototype**
KiCad E.D.A. 10.0.8	Id: 7/7